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Resource and Time-Adjusted Research Productivity Index for Funding Allocations Using Critical Thinking & Ethics in Decision Making

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Abstract

Performance-based funding rewards citations and grants already accumulated, not output relative to the resources a researcher actually had. This paper critiques that logic, using Sorzano and Pueche-Granados (2026) as the test case, and proposes the Resource- and Time-Adjusted Research Productivity Index (RTA-RPI): scientific output divided by cumulative resource exposure. RTA-RPI exposes funding efficiency that raw bibliometrics hide, and gives early-career and underfunded researchers a fair basis for comparison. Applied math is only as fair as the assumptions behind it: critical thinking about what a formula leaves out — who had less time, less money, less visibility — matters more than the formula itself, since an unexamined metric just launders existing bias through a number neglecting ethics.

Keywords: Research Funding; Researchers Finance; Scientometrics; Research Ethics; Applied Math; Agentic Computer Simulations

1 Introduction

Funding decisions determine which questions get studied and who gets to study them. Most systems use historical performance — publications, citations, h-index, institutional reputation — as the main signal for who gets funded next.

That signal conflates two different things: how much a researcher has produced, and how much opportunity they had to produce it. A senior professor with decades of multimillion-dollar funding and a researcher with almost none are not comparable on raw output alone.

This is the Matthew effect in science: accumulated advantage compounds, so established researchers who already hold funding, infrastructure, and visibility keep receiving more of each, while early-career researchers and those without substantial initial support fall further behind regardless of underlying ability.[2] A performance-based funding system built on historical output does not correct for this effect; it formalizes it.

Sorzano and Pueche-Granados (2026) propose performance-weighted allocation mechanisms built on exactly this signal.[1] Their framework is useful, but it inherits the conflation: it predicts who will keep producing, not who would produce the most per dollar of opportunity.

Central claim: scientific productivity should be evaluated relative to the resources and time available to generate it — not by accumulated performance alone. The RTA-RPI operationalizes this.

2 The Problem with Performance-Weighted Funding

Prediction ≠ policy. Sorzano and Pueche-Granados show historical performance predicts future performance across ∼4,400 researchers.[1] That does not establish that funding the predicted winners is the fairest or most valuable use of money — prediction and allocation are different questions.

No resource denominator. Their performance signal is never divided by what it cost to produce. A researcher with $10M and 20 years is compared to one with $5K and 2 years on the same scale, which rewards access to money as if it were skill.

Concentration. Their own stochastic model, optimized for impact alone, converges to funding just 5 of ∼4,400 researchers. The authors admit fairness and diversity must be imposed as separate constraints — they don't emerge from the objective. An allocation can be mathematically optimal and still inequitable.

Early-career penalty. Qualified new researchers must out-compete decades of institutional advantage just to get their first independent funding. Historical performance shouldn't be the sole gate to that opportunity.

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3 The RTA-RPI

Let P_i be researcher i's field-normalized scientific performance (initially: citations), $R_i(t)$ their annual accessible research resources, and T_i their active research period. Cumulative resource exposure (Eq. 1) is defined as:

$$E_i = \int_0^{T_i} R_i(t)\, dt \tag{1}$$

When a researcher has documented career breaks (parental leave, illness, or other interruptions), exposure is instead computed discretely over active periods only (Eq. 2):

$$E_i = \sum_{t=1}^{T_i} R_i(t)\, a_i(t) \tag{2}$$

where $a_i(t) = 1$ if researcher i was active in year t, and 0 during a documented break — so break years contribute zero exposure rather than diluting the researcher's average.

The index itself (Eq. 3) is performance per \$1,000 of cumulative exposure:

$$RTA\text{-}RPI_i = \frac{P_i}{E_i} \times 1000 \tag{3}$$

When annual resources are roughly constant, Eq. 3 reduces to $P_i \div (R_i \times T_i) \times 1000$.

A higher score means more output per dollar of opportunity — it does not by itself prove higher quality, or that more funding would scale proportionally. It adds a dimension absolute performance rankings omit.

4 Illustrative Examples

Ten hypothetical researchers, same field-normalized citation basis illustrated in Table 1. Each case's Exposure and RTA-RPI value was computed with a simple agentic computer simulation that applied Eq. 1–3 directly to the inputs below, rather than by hand:

With 10 profiles, the RTA-RPI spread runs from 0.044 (Case G) to 25.0 (Case J) — a ∼570× range, while raw citations only range ∼200× (50 to 10,000). Case G and Case J anchor the two extremes and are the clearest cases to explain the index by. Three patterns stand out:

Scale breaks the raw-citation ranking. Case G has the most citations (10,000) of anyone in the set but the *lowest* RTA-RPI (0.044) — \$225M bought that output, 2.5× Case A's exposure for 2.5× Case A's citations, so efficiency is essentially flat between them despite Case G looking far more "productive" on paper. Case G is the model's cautionary example: raw scale without a resource denominator overstates productivity.

Time dilutes the index, not just money. Case B and Case J share the same near-zero funding rate (\$5K and \$2K/yr) but Case J's single-year window gives it the highest score in the set (25.0) — short, cheap, high-output stints score best; Case F and Case C show the inverse: real output stretched over decades of low funding (1.6 and 1.33) lands mid-table, not at the top, because years accumulate in the denominator even when dollars don't. Case J is therefore the model's open question: it is the index's top performer and its clearest case for a minimum-exposure safeguard (Section 6), since a single lucky year should not outscore a lifetime of comparable efficiency.

Mid-tier funding does not guarantee mid-tier efficiency. Case D, Case E and Case I all sit in the \$300K–\$2M/yr band but span 0.06–0.67, a >10× difference — Case E's heavy short-term funding is the least efficient of the three; Case I's lower, longer, steadier funding is the most. Funding level alone does not predict RTA-RPI; the citations-per-dollar-year ratio does.

Overall, RTA-RPI and raw citations rank researchers almost oppositely at the extremes (compare Case A and Case G at the bottom vs. Case B and Case J at the top), which is exactly the inversion the index is built to surface.

5 Application to Funding Allocation

RTA-RPI is a funding-*efficiency* signal, not a universal ranking. Decision procedure:

1. **Eligibility gate** — scientific quality, feasibility, integrity, ethics, independent of RTA-RPI.
2. **Resource exposure** — estimate cumulative accessible funding per researcher over the evaluation period.
3. **Compute RTA-RPI** — field-normalized performance ÷ verified exposure.
4. **Weight toward opportunity** — give particular consideration to high-RTA-RPI researchers where more funding would expand independent capacity.
5. **Protect early-career access** — a funding pathway that does not force new researchers to out-compete lifetime citation totals. RTA-RPI supports this; it should not become a new mandatory leaderboard.

6 Methodological Safeguards

- **Disciplinary normalization** — citation patterns differ by field; use field-normalized, publication-year-adjusted measures (Leiden Manifesto; DORA 2024).[3, 4]
- **Resource attribution** — separate a PI's direct individual budget from shared infrastructure, personnel, and collaborative resources; attribute transparently.
- **Inflation adjustment** — convert all historical funding to constant purchasing power.
- **Career interruptions** — adjust the active period for documented leave or illness; measure actual research time, not elapsed calendar time.
- **Citation attribution** — use contribution-based, not full, credit on collaborative work.
- **Citation lag** — use appropriate citation windows so recent work isn't penalized for having less time to accumulate citations.
- **Minimum exposure threshold** — cap or flag extreme RTA-RPI values from very low or undocumented exposure; report as uncertain, not exact.

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Researcher	Citations	Annual resources	Years	Exposure	RTA-RPI
Case A — senior professor	4,000	$10M	20	$200M	0.020
Case B — early-career	200	$5K	2	$10K	20.000
Case C — long-term underfunded	200	$5K	30	$150K	1.333
Case D — mid-career, moderate funding	1,500	$500K	10	$5M	0.300
Case E — richly funded, short career	600	$2M	5	$10M	0.060
Case F — underfunded, long career	800	$20K	25	$500K	1.600
Case G — superstar, maximal funding	10,000	$15M	15	$225M	0.044
Case H — early-career, modest funding	350	$50K	3	$150K	2.333
Case I — efficient workhorse, mid funding	3,000	$300K	15	$4.5M	0.667
Case J — minimal funding, minimal time	50	$2K	1	$2K	25.000

Table 1: Illustrative RTA-RPI comparison across ten simulated researcher profiles.

6.1 Ethical Considerations

Any index that moves real money carries ethical weight beyond its math. Six concerns apply directly to RTA-RPI:

- **Gaming risk** — if exposure self-reporting is unverified, researchers have an incentive to understate funding to inflate their score; verification by funders, not self-report, is required.
- **Due process** — a low RTA-RPI must never be treated as evidence of low ability; it should only ever narrow or widen opportunity, never serve as public judgment of a researcher's worth.
- **Equity vs. merit tension** — correcting for resource inequality is a fairness choice, not a neutral technical one; it should be adopted transparently as policy, not smuggled in as if the formula alone settles what funders owe each other.
- **Citations without context is a double failure** — a senior researcher citing raw citation counts while ignoring decades of funding and time behind them is not just ethically unfair to those with less access; it is mathematically weak, since an unadjusted number conflates opportunity with ability and so proves less than it appears to.
- **Academic exploitation and bullying** — failure to account for indexes like RTA-RPI leaves resource-adjusted output invisible to institutions, which lets senior researchers over-pressure early-career researchers, including postdoctoral researchers and PhD-level students, into producing under funding and time conditions that would be recognized as inequitable if measured. This dynamic is the specific subject of a parallel case study on postdoctoral exploitation tactics in AI-era academia.[10]
- **Self-citation adjustment** — under the simple form of RTA-RPI, citations a researcher makes to their own prior work should be subtracted from P_i before the index is computed, so self-citation cannot inflate the numerator; this revised scoring keeps RTA-RPI aligned with externally recognized contribution rather than self-generated citation count.

7 Limitations

- **Not a causal claim.** A high RTA-RPI may reflect personal effort, collaboration, or access to tools built by others, not funding efficiency alone; the index measures output relative to resources, but does not prove that the resources caused the output.
- **Citations $\neq$ quality.** Citation count says nothing about originality, reproducibility, or real-world impact, so a high RTA-RPI does not mean the underlying research is better — only that it was produced with less money and time.
- **Funding buys more than citations.** Grants also fund infrastructure, train doctoral students, and transfer technology to industry; none of those outcomes are counted by RTA-RPI, which only tracks citation-based performance.
- **Resource exposure is hard to isolate.** On large, multi-investigator grants, splitting shared funding fairly between individual researchers is genuinely difficult; an inaccurate exposure estimate distorts the index directly, since exposure sits in the denominator.
- **Not yet tested on real data.** The illustrative cases in Section 4 are simulated, not observed; RTA-RPI still needs validation against actual longitudinal funding and citation records before a funder should rely on it.

8 Conclusion

Accumulated performance is not the same as efficient use of opportunity. Sorzano and Pueche-Granados show that impact-maximizing allocation concentrates funding in the already-funded — a mathematically sound outcome that is not necessarily an equitable one. RTA-RPI reframes the question: not how much has a researcher produced, but how much per dollar of opportunity they actually had. Scientific funding should reward contribution and resource efficiency — not simply finance the continuation of accumulated advantage.

Acknowledgments

This paper was written with AI assistance and Claude models built by Antropic company. The illustrative Case A–J comparisons in Section 4 were produced by simple agentic computer simulations that applied the RTA-RPI formulas derived in Section 3 to different researcher input values; no empirical dataset was used.

Conflict of Interest

The author declares no conflict of interest.

Author Contributions

This paper has one human author (D.S.), who conceived the index, designed the illustrative simulations, and prepared the manuscript.

Data Availability Statement

All illustrative data in this paper are simulated for demonstration purposes and are reproducible from the formulas given in Section 3. No empirical dataset was used.

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References

[1] Sorzano, C. O. S., & Pueche-Granados, B. (2026). *Research Funding as a Decision Problem Under Heavy-Tailed Uncertainty*. arXiv preprint, arXiv:2604.22793. https://doi.org/10.48550/arXiv.2604.22793

[2] Merton, R. K. (1968). The Matthew Effect in Science. *Science*, 159(3810), 56–63. https://doi.org/10.1126/science.159.3810.56

[3] Hicks, D., Wouters, P., Waltman, L., de Rijcke, S., & Rafols, I. (2015). Bibliometrics: The Leiden Manifesto for research metrics. *Nature*, 520, 429–431. https://doi.org/10.1038/520429a

[4] San Francisco Declaration on Research Assessment (DORA). (2024). *Guidance on the Responsible Use of Quantitative Indicators in Research Assessment*. https://sfdora.org/resource/guidance-on-the-responsible-use-of-quantitative-indicators-in-research-assessment/

[5] Wilsdon, J., Allen, L., Belfiore, E., Campbell, P., Curry, S., Hill, S., Jones, R., Kain, R., Kerridge, S., Thelwall, M., et al. (2015). *The Metric Tide: Report of the Independent Review of the Role of Metrics in Research Assessment and Management*. https://doi.org/10.13140/RG.2.1.4929.1363

[6] Seglen, P. O. (1997). Why the impact factor of journals should not be used for evaluating research. *BMJ*, 314, 498–502. https://doi.org/10.1136/bmj.314.7079.497

[7] Radicchi, F., Fortunato, S., & Castellano, C. (2008). Universality of citation distributions: Toward an objective measure of scientific impact. *Proceedings of the National Academy of Sciences*, 105(45), 17268–17272. https://doi.org/10.1073/pnas.0806977105

[8] Aagaard, K., Kladakis, A., & Nielsen, M. W. (2020). Concentration or dispersal of research funding? *Quantitative Science Studies*, 1(1), 117–149. https://doi.org/10.1162/qss_a_00017

[9] Gross, K., & Bergstrom, C. T. (2019). Contest models highlight inherent inefficiencies of scientific funding competitions. *PLOS Biology*, 17(1), e3000065. https://doi.org/10.1371/journal.pbio.3000065

[10] Saatchi, D. (2026). *Researchers Finance Comedy: Excellent Innovative Strategies to Exploit Postdoctoral Researchers in AI Era*. AI Journal of Arts and Metascience (AIJAM), Zenodo. https://doi.org/10.5281/zenodo.22060258